

Day - 2: Express and APIs in Node.js

Why We Don't Use HTTP Module Directly

- Verbose and repetitive.
 - No routing, middleware, or extra features.
 - Hard to manage large applications.
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Introduction to Express

- Lightweight, fast Node.js framework.
 - Simplifies server creation and routing.
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Create a Server Using Express

```
npm install express
```

```
const express = require('express');  
const app = express();
```

Express Creates the Server But Doesn't Start It

- `express()` creates an instance of the app.
 - The server doesn't listen until you call `app.listen()`.
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Start the Express Server

```
app.listen(PORT, () => {  
  console.log(`Server running on port ${PORT}`);  
});
```

! Make the First Request

- Visit `http://localhost:PORT/`
- No API defined yet → shows **Cannot GET /**

Explanation of APIs in Express

- Define routes using:

```
app.get(path, callback);
```

- Each route handles a specific HTTP request.

Create a / API

```
app.get('/', (req, res) => res.send('Hello World'));
```

What is `req` (Request)?

- Object containing client request data:
 - URL
 - Headers
 - Query parameters
 - Body data, etc.

What is `res` (Response)?

- Object used to send data back to client.
- Common methods:
 - `res.send()`
 - `res.json()`
 - `res.status()`

What is an API?

- **API (Application Programming Interface)**: A way for two apps to **communicate**.
 - Allows one software to request data or services from another.
 - Flexible in communication structure (no strict format).
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What are REST APIs?

- **REST** = Representational State Transfer.
 - A set of **rules** for designing APIs.
 - Uses HTTP methods:
 - `GET` , `POST` , `PUT` , `DELETE`
 - Follows standard request/response structures.
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Ways to Send Data in Requests

1. `req.body`

- Sent in body of request.
- Used in `POST` , `PUT` , `PATCH` .
- Example: Form data or JSON.

2. `req.query`

- Data in URL after `?` as key-value pairs.
- Mostly used in `GET` .
- Example:

```
GET /search?city=delhi&limit=10
```

3. `req.params`

- Data in URL path.
- Example:

```
GET /user/ankur_bit_io
```

- Accessed as:

```
req.params.username
```

Request-Response Lifecycle

1. **Client sends a request** to server
2. **Server receives** the request
3. **Route handler matches** the request
4. **Processes** the request
5. **Sends back a response** to client
6. **Client receives** the response Displaying D116 md